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GRADUATION PROJECT

DEVELOPMENT OF A MULTIMODAL 3D OBJECT RECOGNITION AND VISUALIZATION SYSTEM FOR TRAFFIC ENVIRONMENTS USING CAMERA AND LIDAR FUSION

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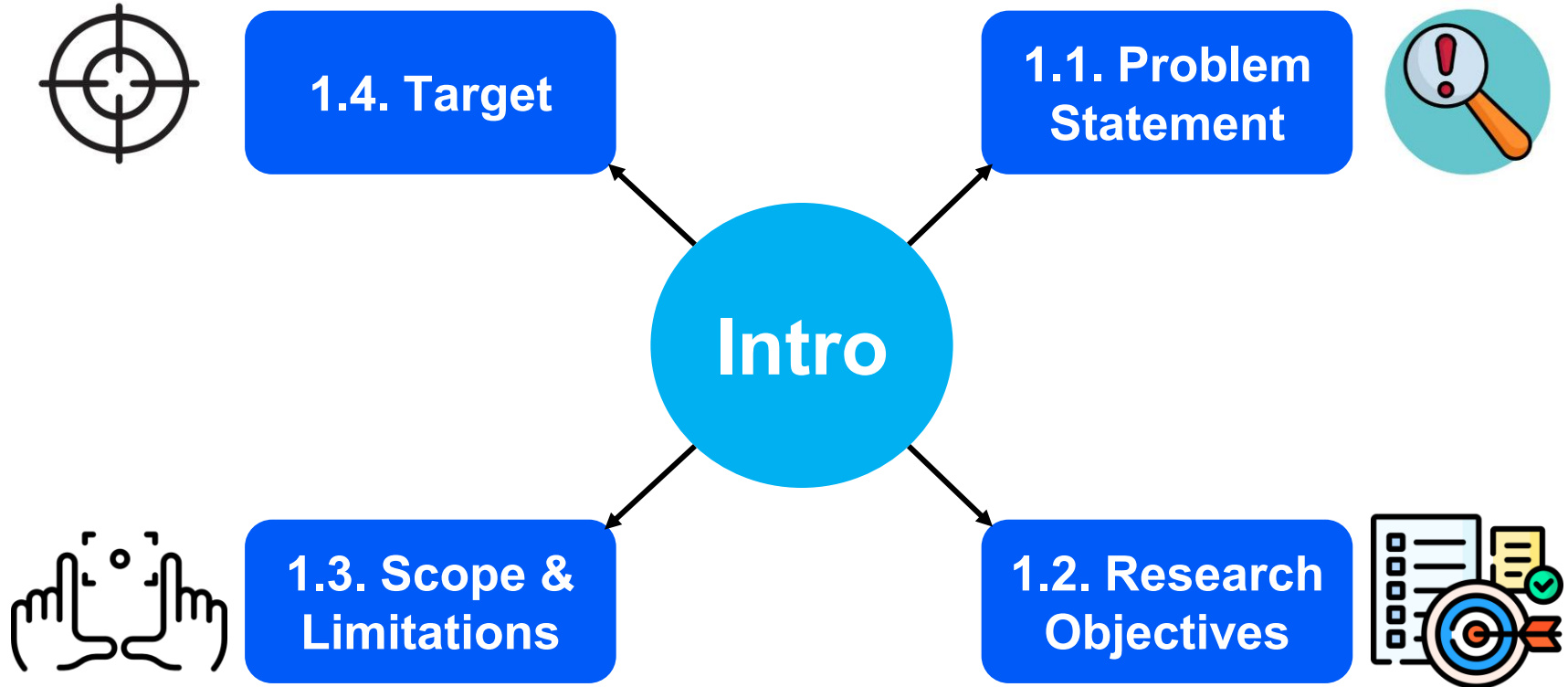
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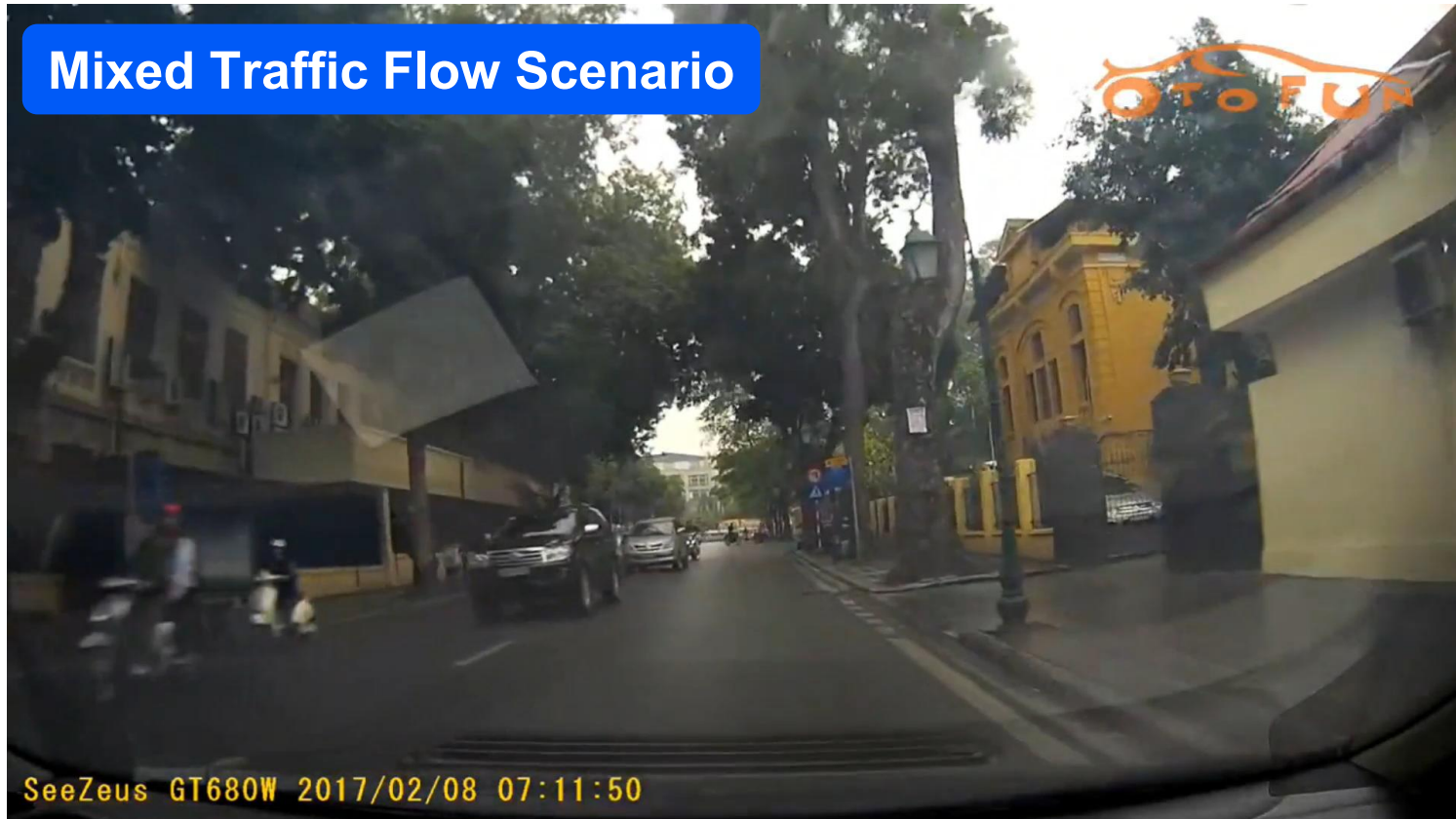
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1. INTRODUCTION



1.1. PROBLEM STATEMENT

Mixed Traffic Flow Scenario



1.1. PROBLEM STATEMENT

Low Lighting Scenario

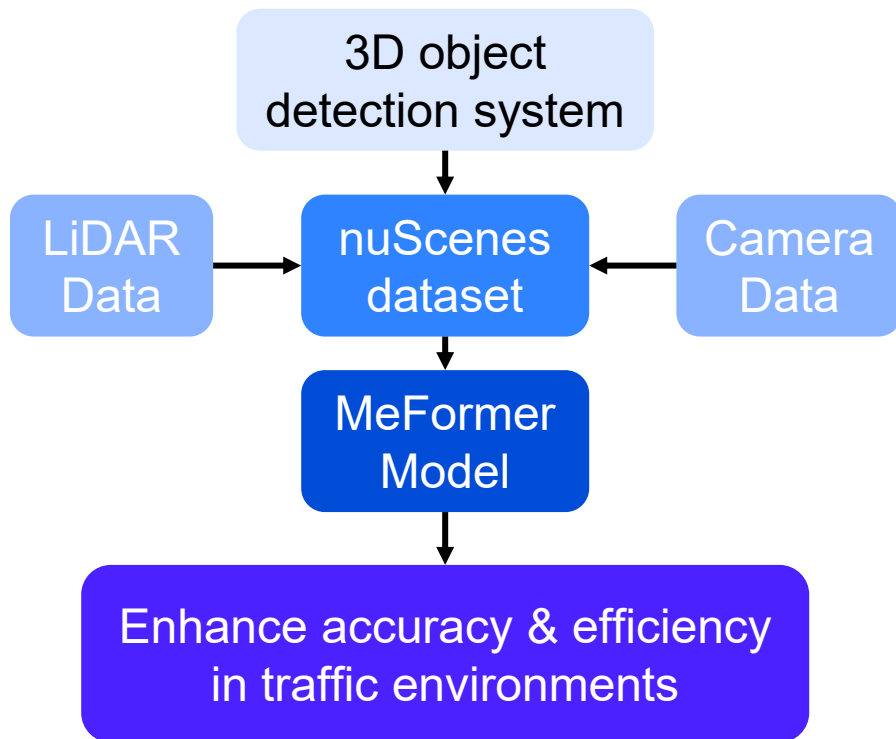
Warning

Some viewers may find the following footage distressing

**The
Guardian**

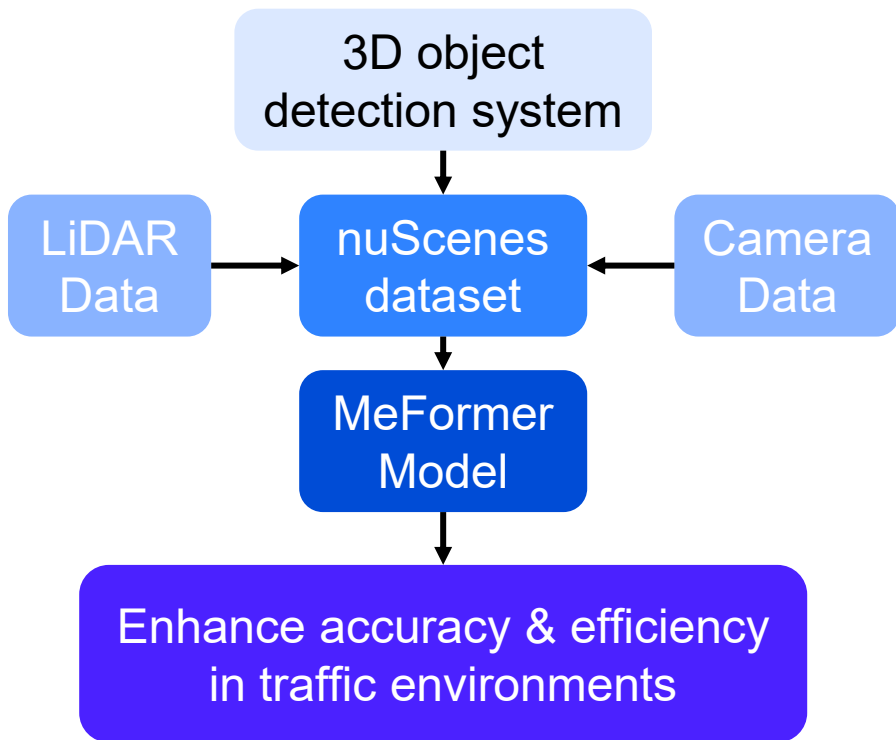
1. INTRODUCTION

1.2. RESEARCH OBJECTIVES

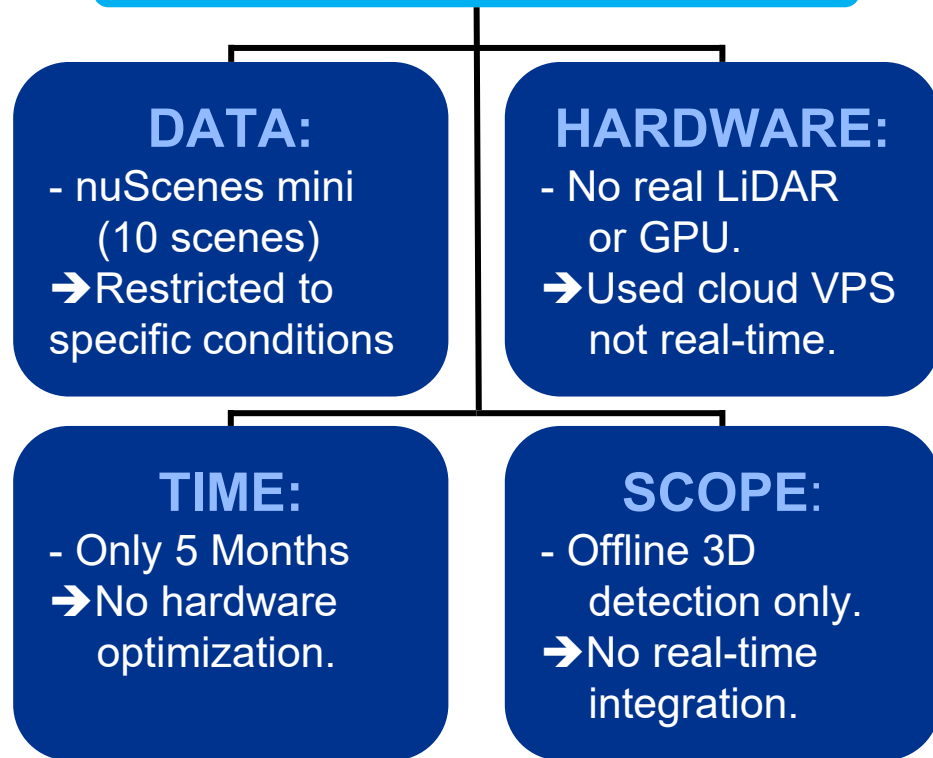


1. INTRODUCTION

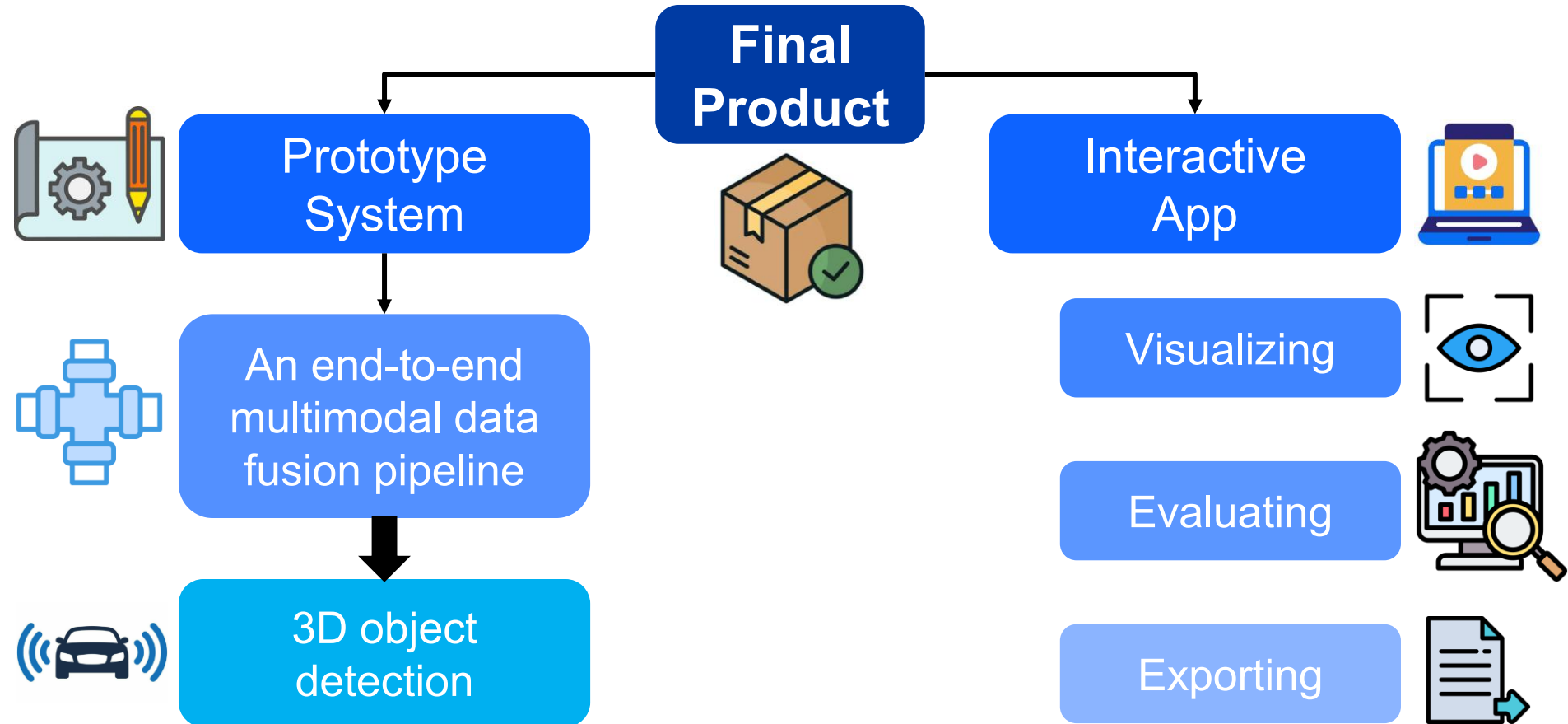
1.2. RESEARCH OBJECTIVES



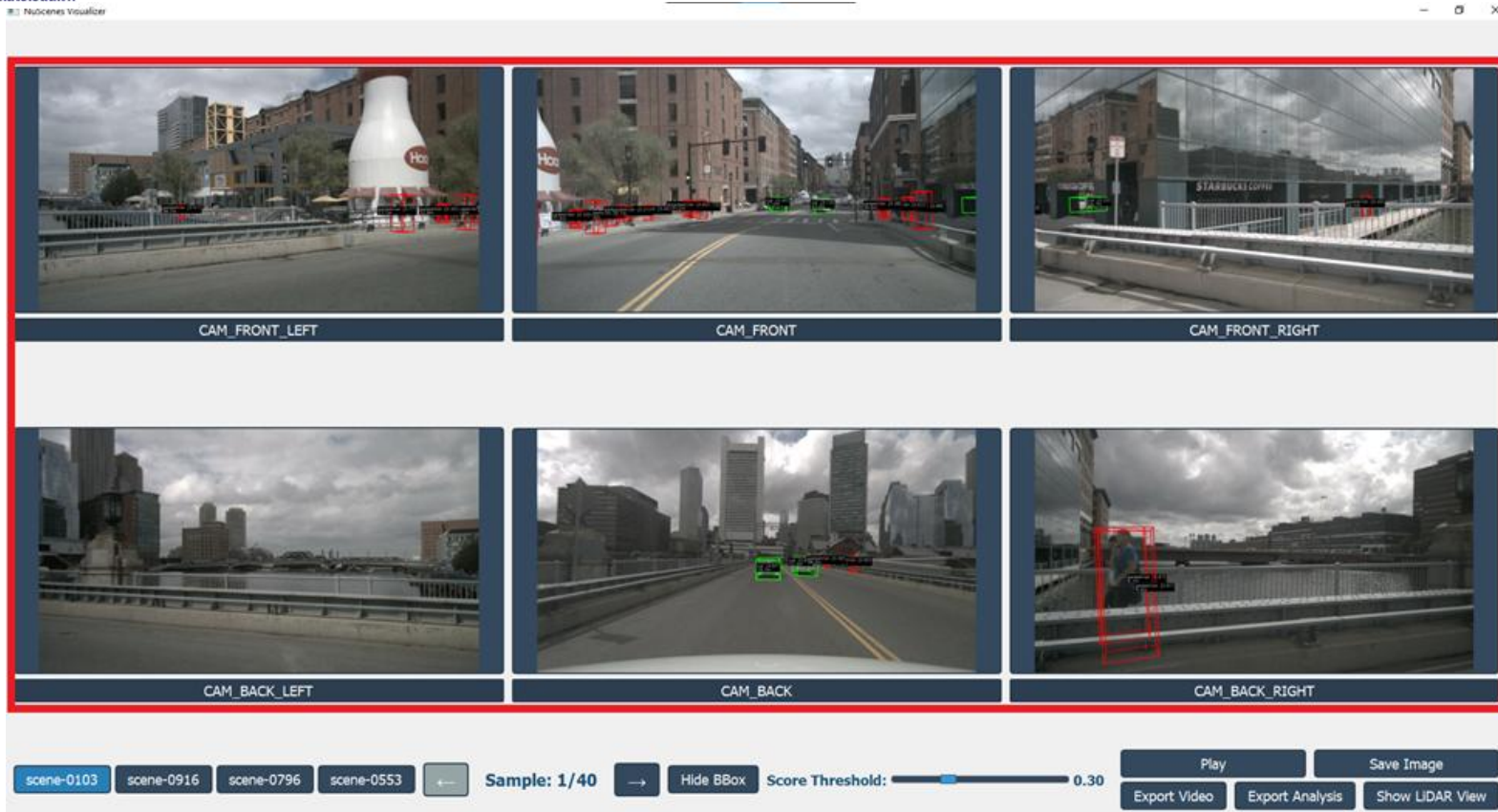
1.3. SCOPE & LIMITATIONS



1.4. TARGET

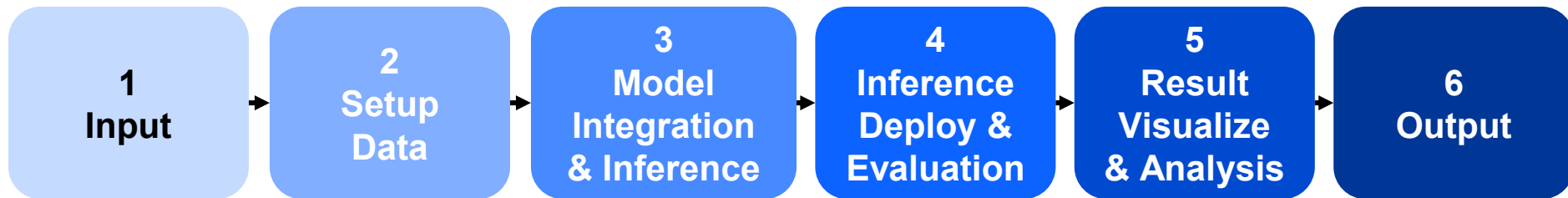


1.4. TARGET



Main User Interface of the Application

2. OVERVIEW OF PIPELINE



2. OVERVIEW OF PIPELINE

- Camera images
- LiDAR point cloud
- Calibration parameters

1
Input

2
**Setup
Data**

3
**Model
Integration
& Inference**

4
**Inference
Deploy &
Evaluation**

5
**Result
Visualize
& Analysis**

6
Output

2. OVERVIEW OF PIPELINE

- Camera images
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- Standardize the format for MEFormer
- Integrate images, point clouds & calibration

2. OVERVIEW OF PIPELINE

- Camera images
- LiDAR point cloud
- Calibration parameters

- Detect 3D bounding boxes & labels with MEFormer
- Apply Non-Max Suppression & filter results

- Standardize the format for MEFormer
- Integrate images, point clouds & calibration

1
Input

2
**Setup
Data**

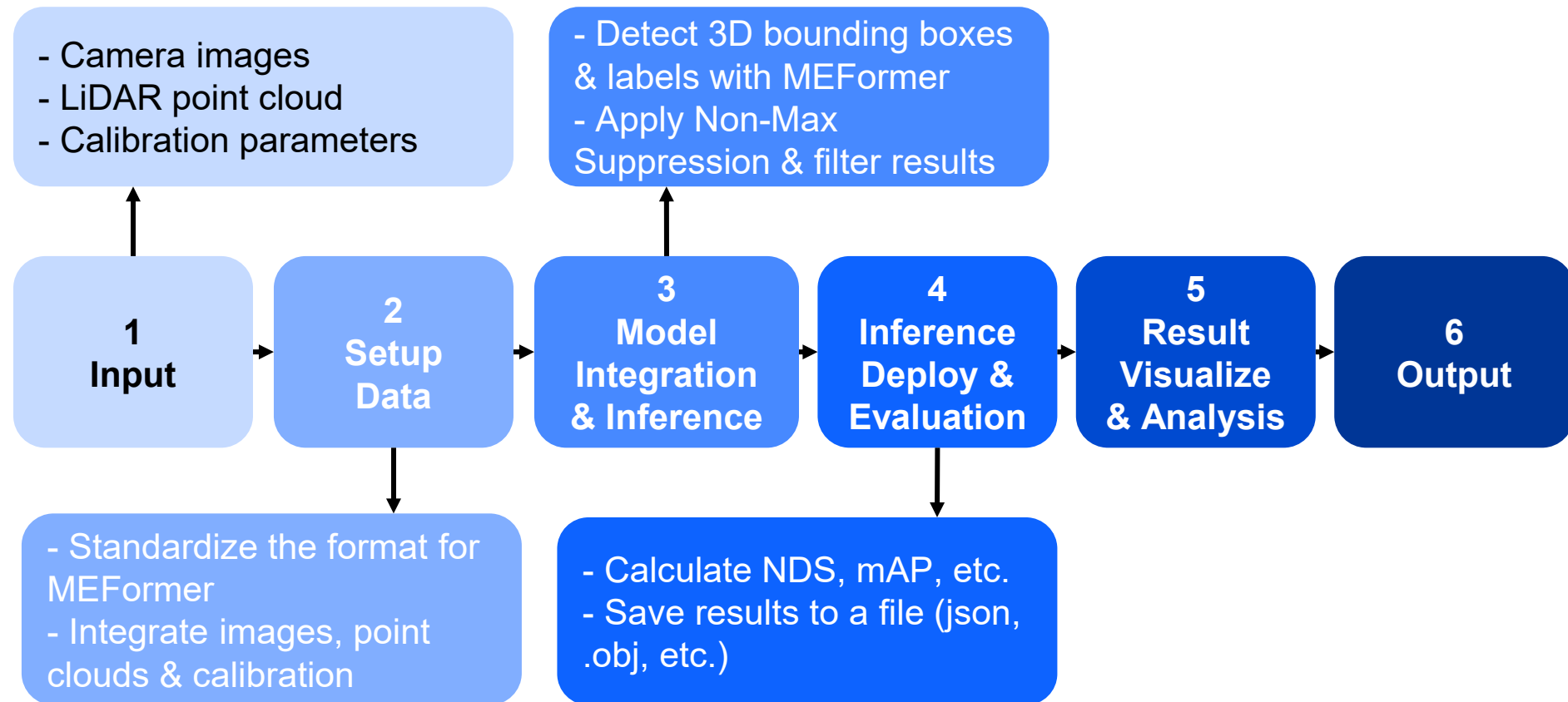
3
**Model
Integration
& Inference**

4
**Inference
Deploy &
Evaluation**

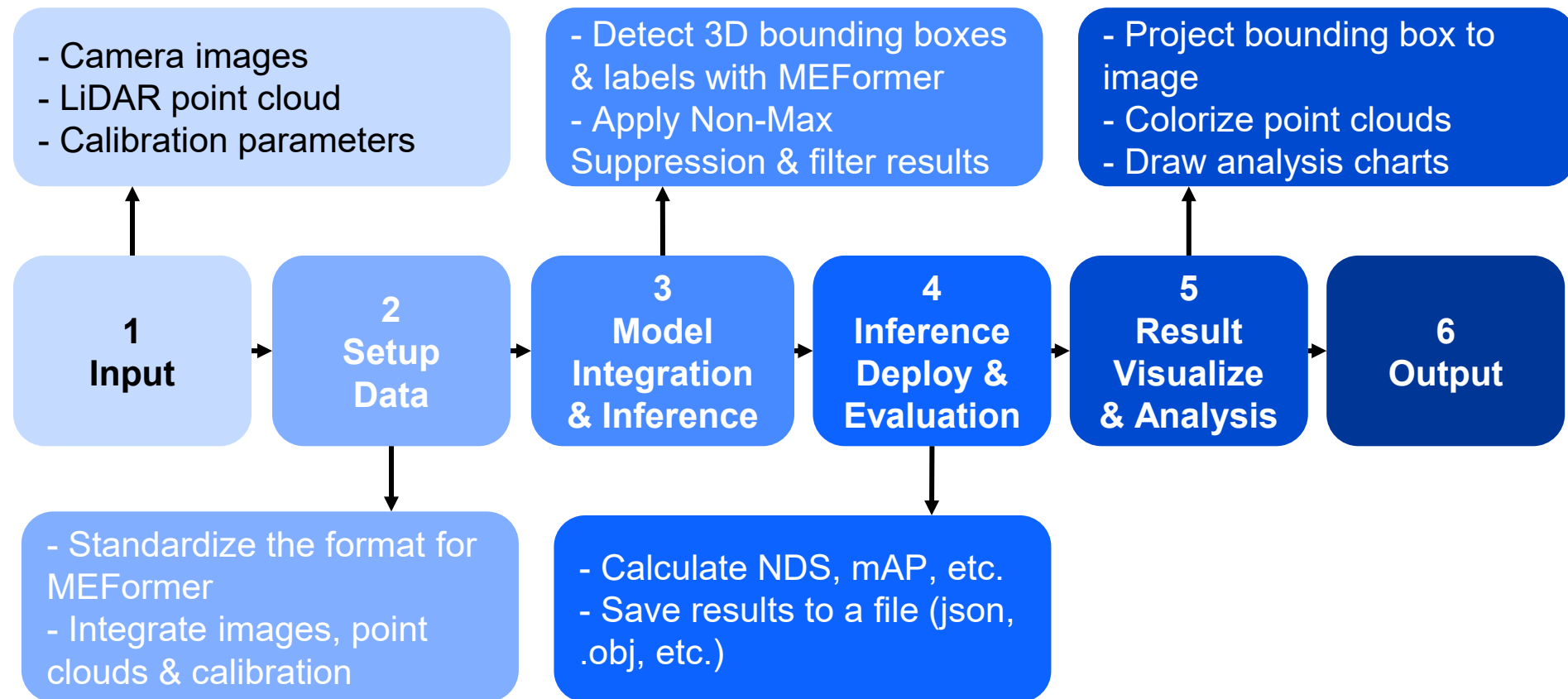
5
**Result
Visualize
& Analysis**

6
Output

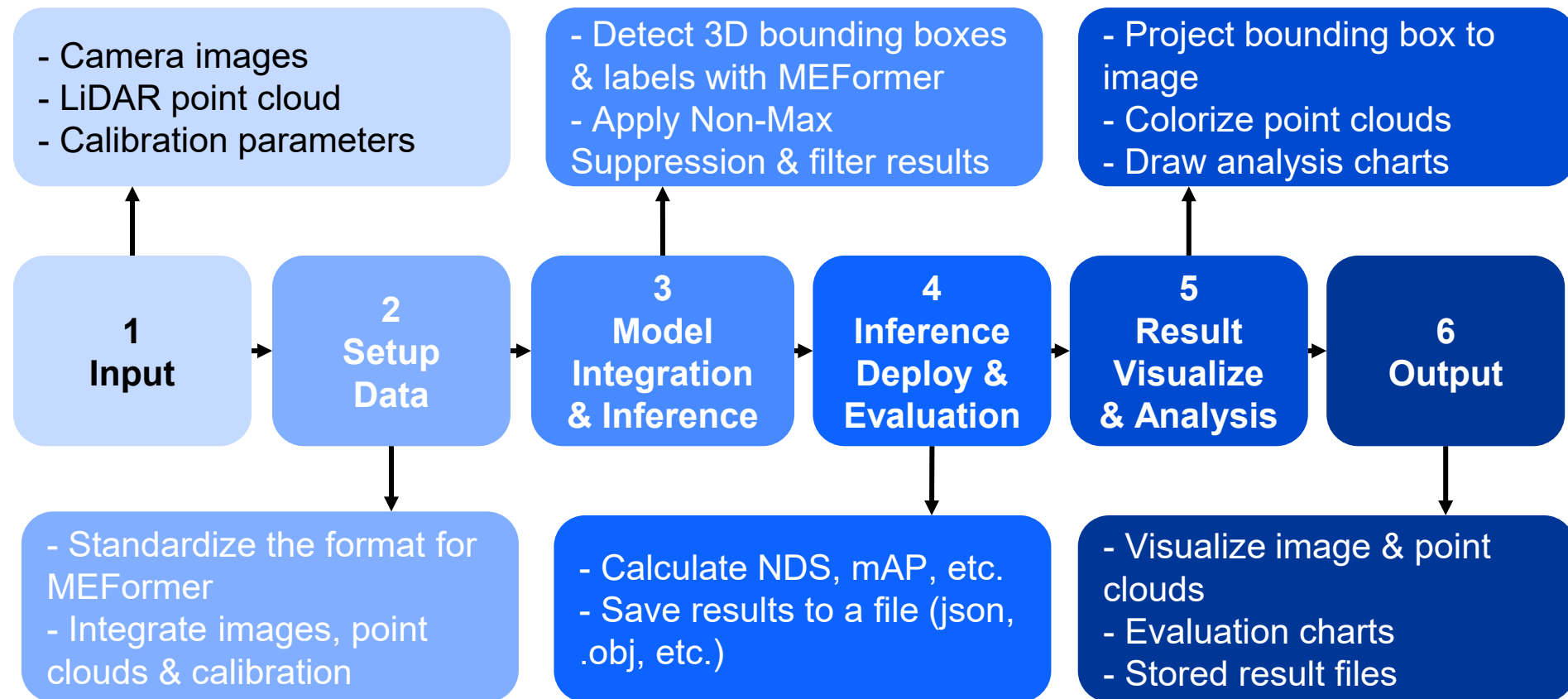
2. OVERVIEW OF PIPELINE



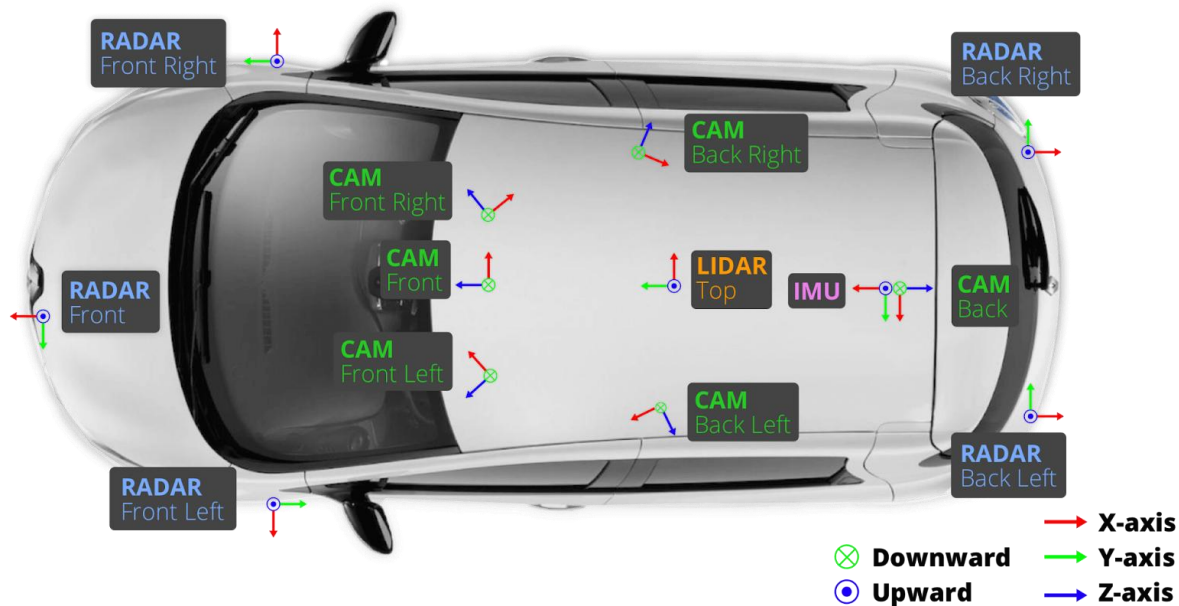
2. OVERVIEW OF PIPELINE



2. OVERVIEW OF PIPELINE



2.1. INPUT & SETUP DATA



Autonomous Vehicle Sensor Layout
in nuScenes Dataset

6 CAMERAS

- Front
- Front-left
- Front-right
- Rear
- Rear-left
- Rear-right

1 LIDAR

- A 32-channel device

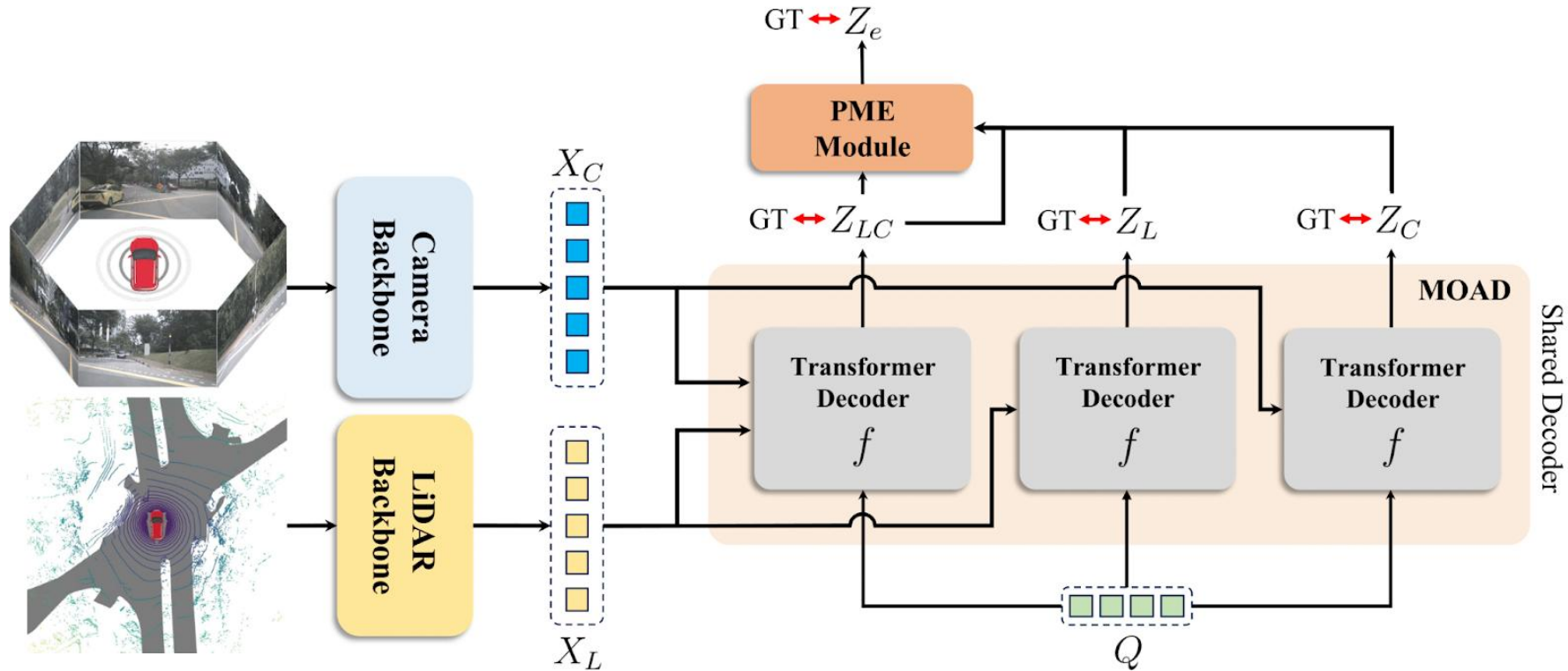
5 RADARS

- Front & 4 Edges

1 IMU

- On the roof

2.2. MODEL INTEGRATION & INFERENCE



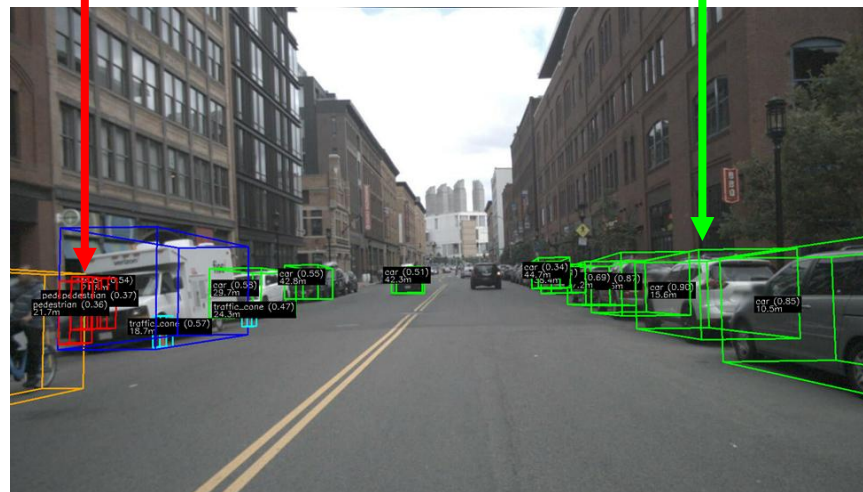
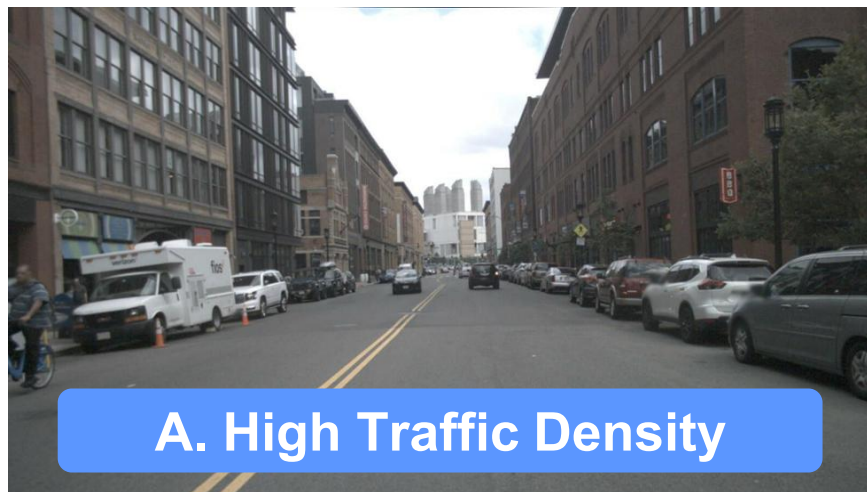
The overall architecture of MEFormer

2.2. MODEL INTEGRATION & INFERENCE

MODELS	mAP (%)	mATE (m)	mAOE (rad)
BEVFormer (C)	44.5	0.58	0.38
DETR3D (C)	41	0.64	0.39
BEVFusion (L+C)	71.3	0.25	0.36
MEFormer (L+C)	72	0.27	0.3
Expected Metrics	≥ 53	≤ 0.49	≤ 0.38
Result	74	0.27	0.21

Performance Comparison of 3D Object Detection Models

2.3. EVALUATION AND VISUALIZATION



Scene-0103, Sample 34 with threshold > 0.3

2.3. EVALUATION AND VISUALIZATION



Scene-0916, Sample 8 with threshold > 0.3

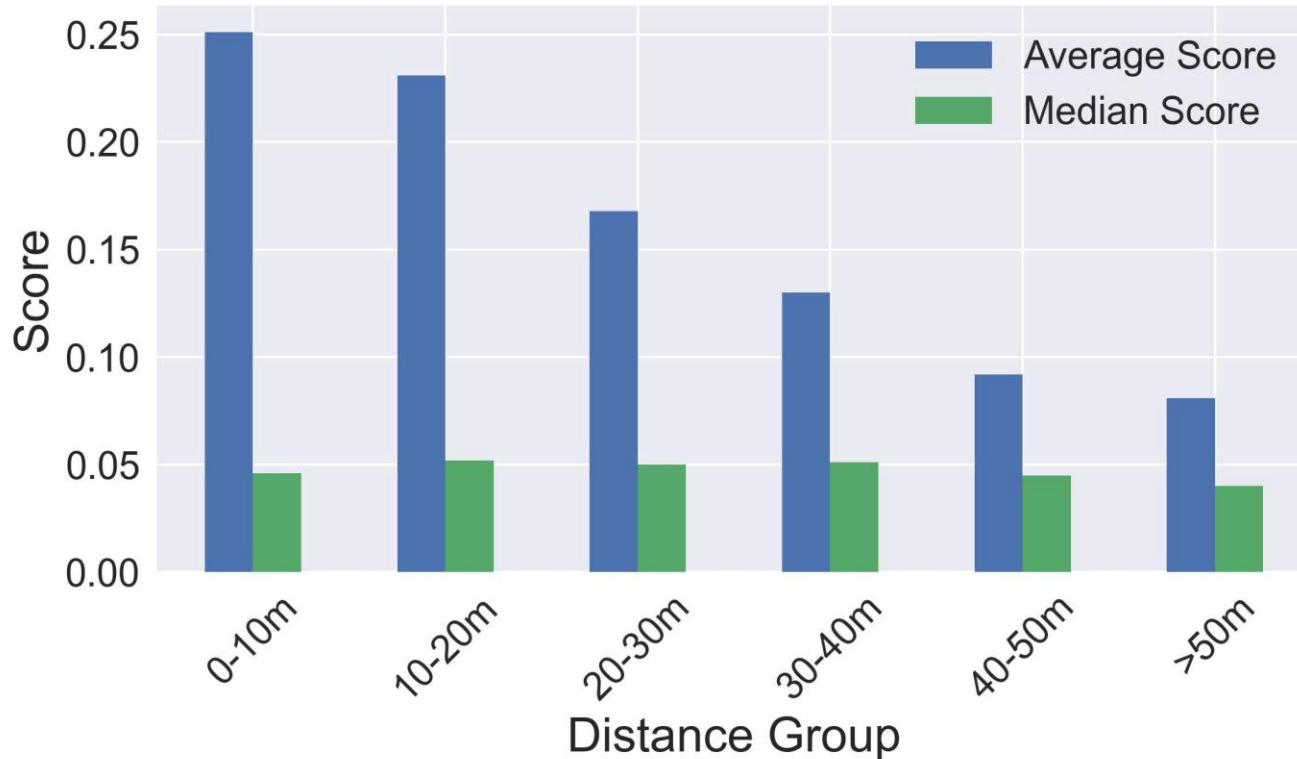
2.3. EVALUATION AND VISUALIZATION



Scene-0796, Sample 11 with threshold > 0.3

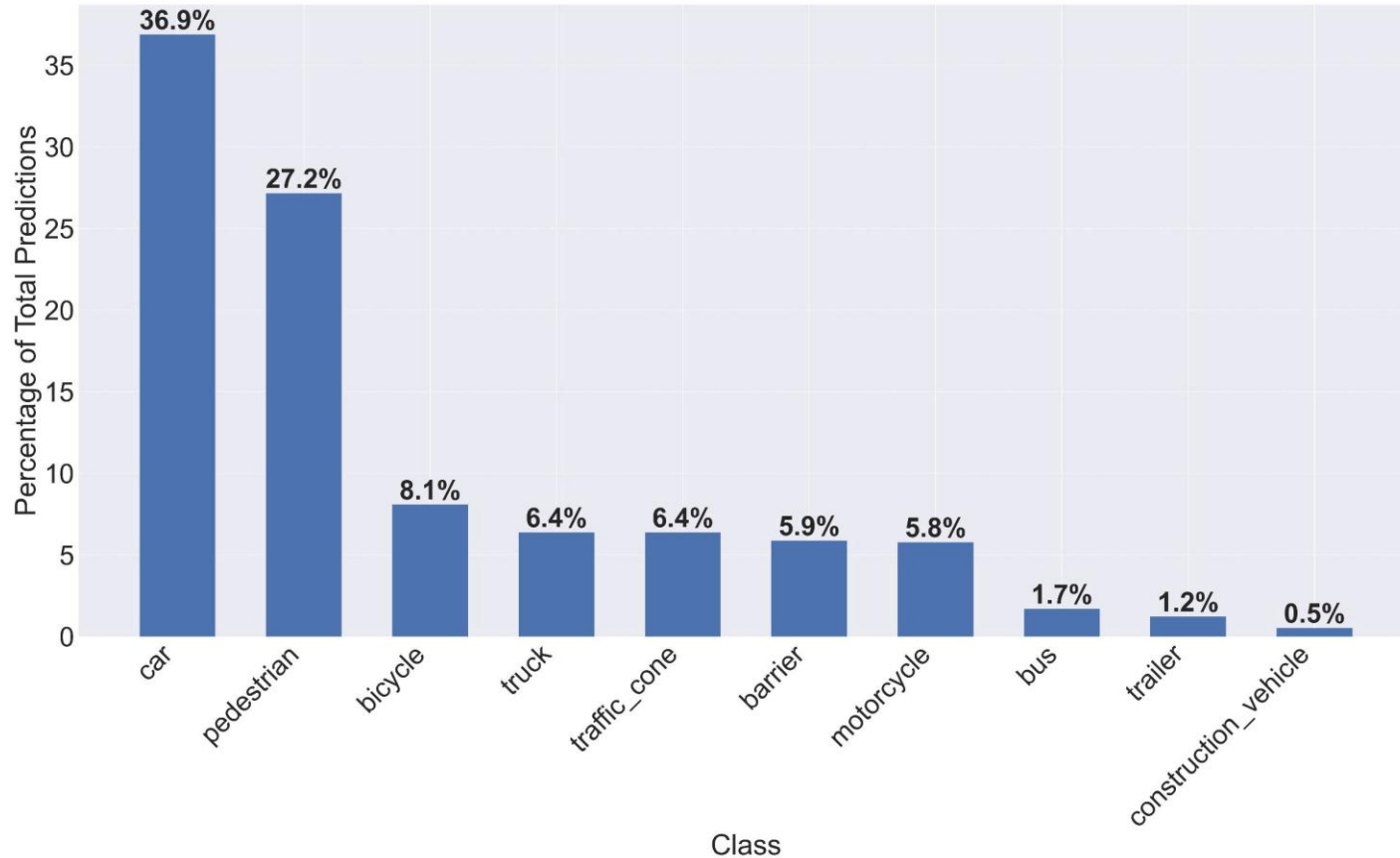
2.3. EVALUATION AND VISUALIZATION

Average and Median Scores by Distance Group



2.3. EVALUATION AND VISUALIZATION

Distribution of Predictions by Class (%)



THE 3-SECOND RULE

A traffic safety guideline for drivers



~2.5-second reaction distance
at 50 km/h (~13.9 m/s)



Recommended Distance = $2.5s \times 13.9m/s = 34.75m$

Our MEFormer model is most reliable under 40 meters



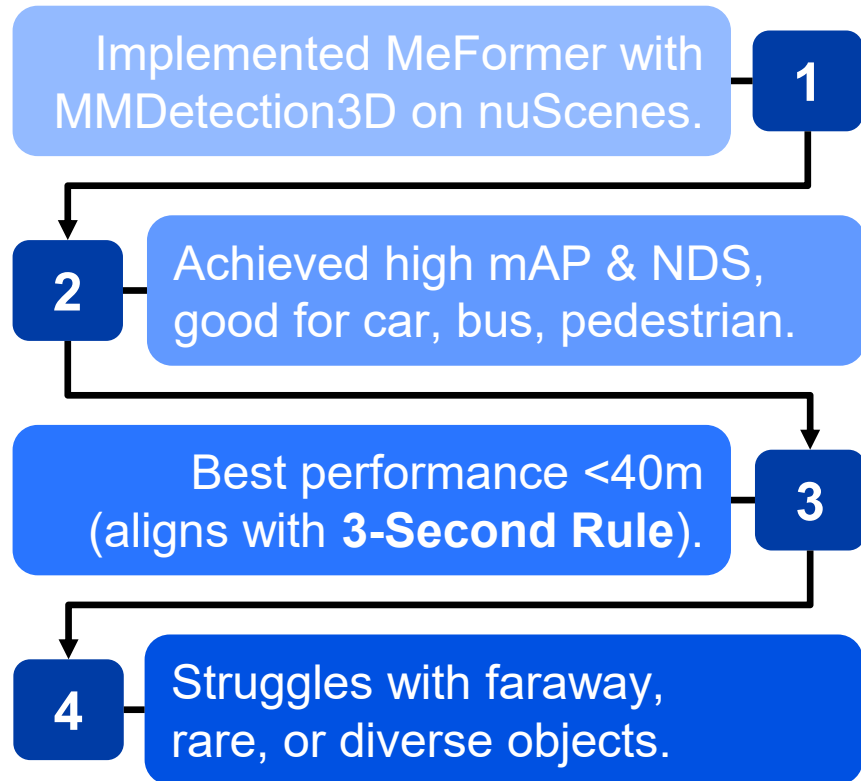
Our MEFormer model meets the 3-Second Rule



DEMO PRODUCT

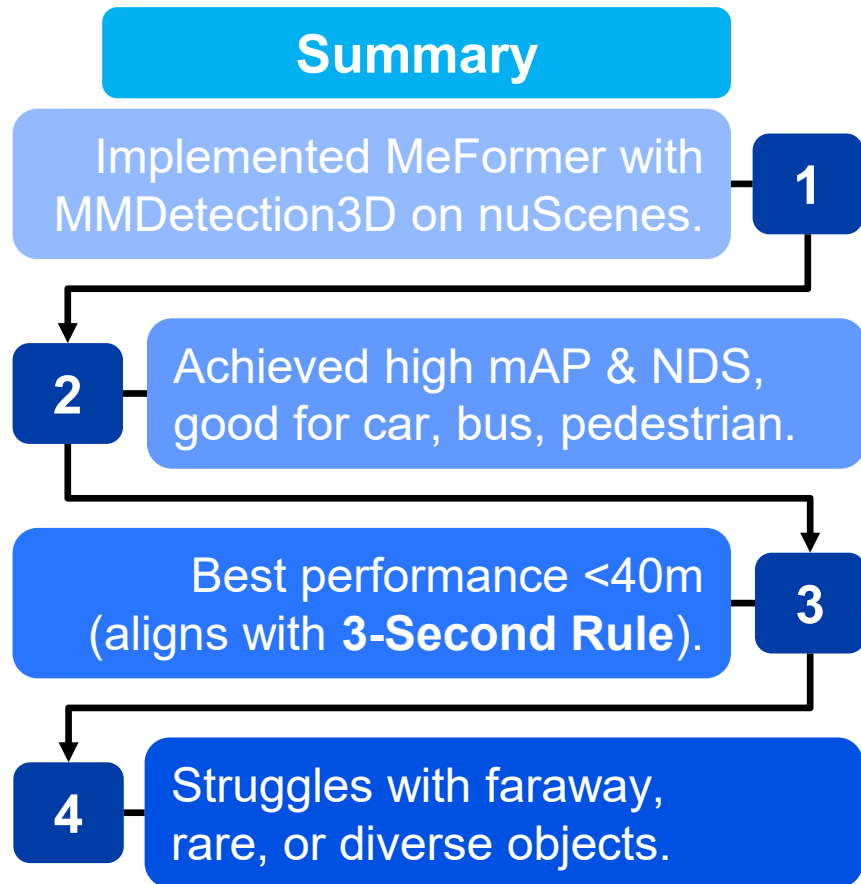
3. CONCLUSION

Summary



3. CONCLUSION

Summary



Future Development

- A HARDWARE SYSTEM:** Build LiDAR/camera system for collecting real-world data.
- B REAL-WORLD TEST:** Deploy on vehicles/simulators to check performance & latency.
- C RARE CLASS BOOST:** Use augmentation/transfer learning for rare/small classes.
- D DYNAMIC THRESHOLDS:** Tune score thresholds by distance or application.



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THANK YOU!

**Development of a multimodal 3D object
recognition and visualization system for traffic
environments using camera and LiDAR fusion**

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